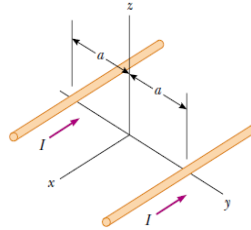
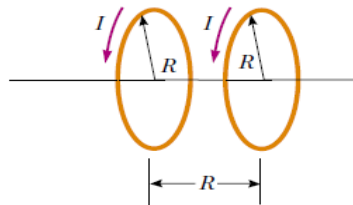


- Two parallel conducting plates are a distance d apart and have a potential difference V between them. A uniform magnetic field B is also present in the region between the plates along a direction parallel to the plates. Particles of charge $+q$ and mass m are released from rest in the crossed E and B fields from the surface of one plate. Find the trajectory of the particles.
- A loop carrying current is shaped like a regular plane polygon of $2n$ sides, the distance between parallel sides being $2a$. Find the magnetic field at the centre of the loop.
- In Figure below, both currents in the infinitely long wires are in the negative x direction. (a) Sketch the magnetic field pattern in the yz plane. (b) At what distance d along the z axis is the magnetic field a maximum?



- Two circular coils of radius R , each with N turns, are perpendicular to a common axis. The coil centers are a distance R apart. Each coil carries a steady current I in the same direction, as shown in Figure below.



- (a) Show that the magnetic field on the axis at a distance x from the center of one coil is

$$B = \frac{N\mu_0 I R^2}{2} \left[\frac{1}{(R^2 + x^2)^{3/2}} + \frac{1}{(2R^2 + x^2 - 2Rx)^{3/2}} \right]$$

- (b) Show that dB/dx and d^2B/dx^2 both are zero at the point midway between the coils. This means the magnetic field in the region midway between the coils is uniform.

- The surface current density in the xoy plane is given to be

$$\vec{K}(x, y) = K_0 \left(\frac{x^2}{a^2} \hat{i} + \frac{y^2}{b^2} \hat{j} \right). \text{ Evaluate } \vec{\nabla} \cdot \vec{K} \text{ and hence calculate the time rate of change of charge contained in the}$$

first quadrant of a circle of unit radius centred on the origin. Show explicitly that for this quadrant, the continuity equation is satisfied in its integral form.

- Consider an infinitely long transmission line consisting of two concentric cylinders having their axes along the z -axis. The cross-section of the transmission line is shown in figure below where z -axis is out of the page. The inner conductor has radius a and carries current I while the outer conductor has inner radius b and thickness d and carries return current $-I$. Determine $\vec{B}$ everywhere, assuming that current is uniformly distributed in both conductors.

